

Distributed Storage

Big Data Systems
School of Data Science
University of Virginia

Last updated: September 16, 2026

Agenda

- > Objects
- > Filesystems
- > Amazon S3
- > HDFS

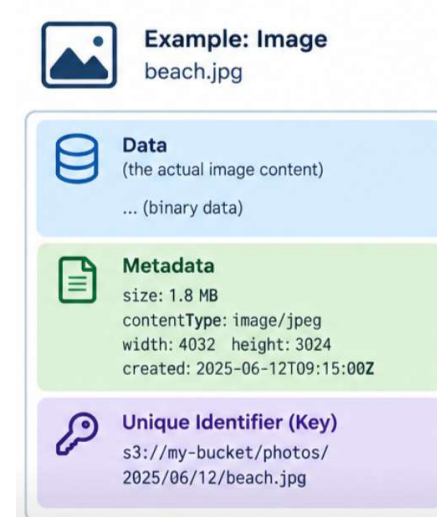
What are Objects?

Objects are fundamental units of data storage in cloud computing

Stored in flat namespace (not in a hierarchical structure)

Each object has three parts:

- > data
- > metadata
- > unique identifier (key)



Objects: Additional Properties

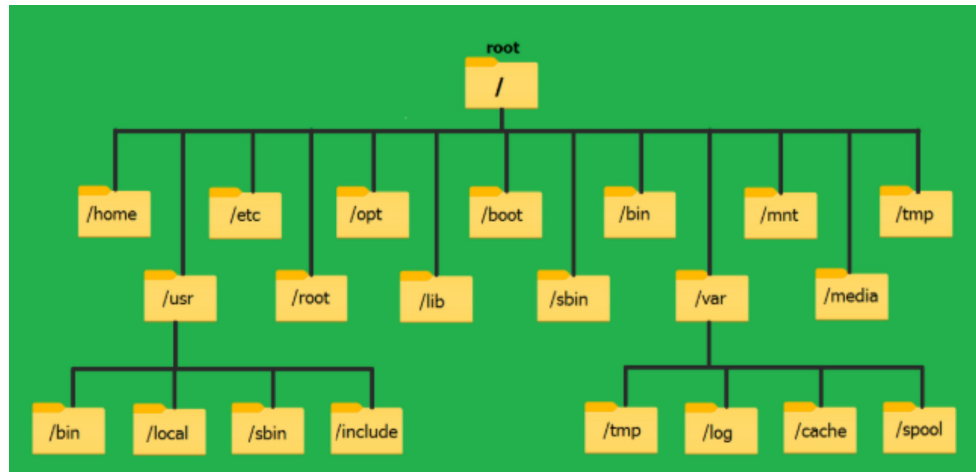
- > Objects are often immutable.
Typically overwrite an object rather than making edit.
- > Since objects are identified by unique keys, they can be spread across multiple servers and data centers

Filesystems

What is a Filesystem?

A filesystem (FS) is the method an operating system uses to organize, store, and retrieve files on a storage device

Here is the Linux FS:



Filesystem Properties

A real filesystem gives you things like:

- > Hierarchical directories (that actually exist)
- > In-place mutation (write at an offset)
- > File locks (to prevent user overwrites and data corruption)
- > Cheap metadata operations

Amazon S3 vs. HDFS

Next, we will look at:

- > Amazon S3, which uses object storage **for scale**
- > HDFS, which uses a distributed filesystem for **throughput**
- > They intentionally offer different user contracts regarding data consistency, performance, and functionality

Amazon S3

Introducing Amazon S3



- > Amazon's managed cloud storage solution
- > Simple Storage Service; not so simple any more
- > Handles durability, replication, scalability behind the scenes
- > Object storage built to store and retrieve any amount of data from anywhere

Amazon S3 Core Ideas

- > Stores data as *objects* in *buckets*
- > Each object has a key (string), metadata, and content
- > Effectively infinite scale
- > No cluster to manage

Amazon S3 Durability

Amazon S3 Standard and most S3 storage classes designed to provide 99.999999999% durability (“eleven nines”) over a year.

What this means:

For every 10,000 objects stored, you'd expect to lose one object on average every 10 million years

However, in Recent News...

 MASS DESTRUCTION

Iran strikes on Amazon data centers caused permanent loss of customer data

War damage to data centers exceeded what AWS services are designed to withstand.

JEREMY HSU – SEP 16, 2026 12:40 PM | 71



→ An Amazon data center in Abu Dhabi, UAE as seen on March 2, 2026. An Iranian drone strike damaged the data center on March 1.
Credit: Planet Labs PBC

Amazon has acknowledged the permanent loss of some customer data that was hosted in Bahrain and the United Arab Emirates.

Amazon S3 Replication

- > S3 is durable thanks to **replication** across multiple data centers and advanced data protection
- > AWS automatically copies data across at least three physically separated Availability Zones within a single region.

Amazon S3 Pricing

Amazon S3

Overview

Features ▾

Storage classes ▾

Pricing

Security

Resources ▾

FAQs

Please note that we list Storage Requests and Data Retrievals Pricing below the Storage Pricing table.

Region:

US East (N. Virginia) ▾

Storage pricing

S3 Standard - General purpose storage for any type of data, typically used for frequently accessed data

First 50 TB / Month

\$0.023 per GB

Next 450 TB / Month

\$0.022 per GB

Over 500 TB / Month

\$0.021 per GB

S3 Intelligent - Tiering * - Automatic cost savings for data with unknown or changing access patterns

Monitoring and Automation, All Storage / Month (Objects > 128 KB)

\$0.0025 per 1,000 objects

<https://aws.amazon.com/s3/pricing/>

Amazon S3 Limitations

- > Higher latency than local disks or HDFS
- > No true filesystem (“Directories” are just key prefixes)
 - Access is via API calls, not file handles
 - Overwrites are whole-object operations
 - Can’t do in-place mutation
 - No file locks

HDFS

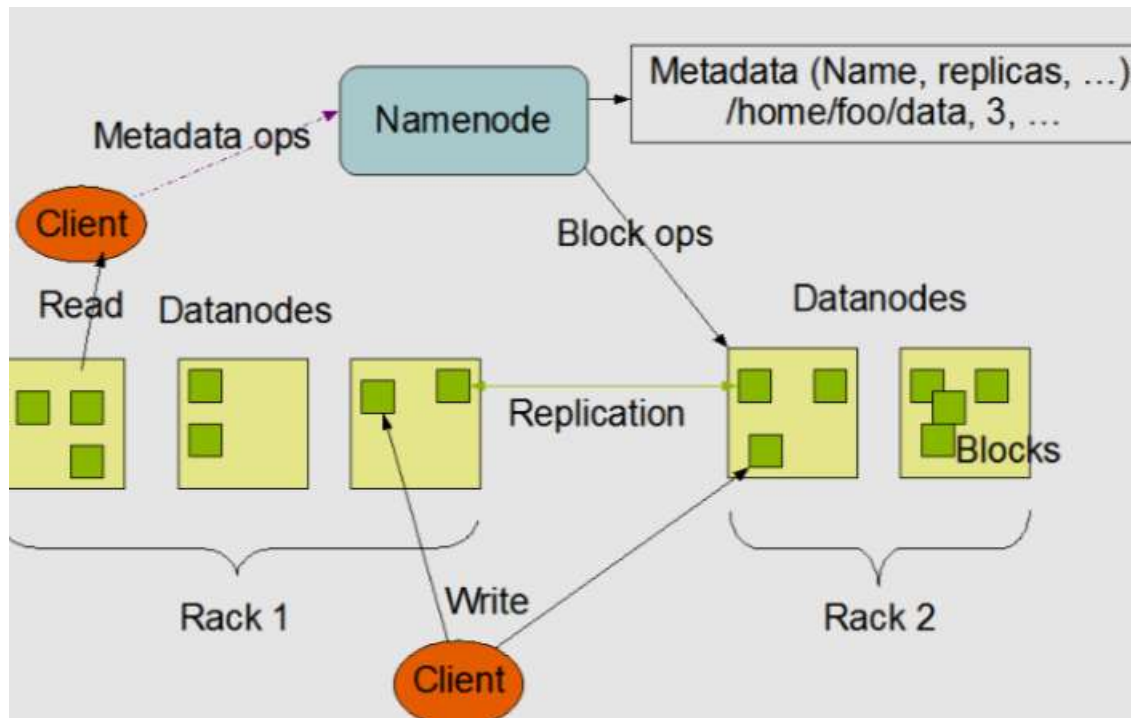
About HDFS

- > Traditional an on-premise solution (not hosted in cloud)
- > Hadoop Distributed File System (HDFS) uses a directory structure
- > Stores very large files (GB–TB scale) across a cluster of commodity machines
- > Built for batch analytics, especially MapReduce / Spark-style workloads

About HDFS

- > Write once, read many (append is allowed; random writes are not)
- > Files split into large blocks (typically 128–256 MB)
- > Each block replicated (3 copies) across different nodes for fault tolerance
- > Optimized for high throughput, not low latency

HDFS Architecture



- > *NameNode* manages metadata (file paths, block locations, permissions)
- > *DataNode* stores blocks of data
- > Client asks Namenode where blocks live, then streams data directly from DataNodes

HDFS Strengths

- > Excellent for large sequential reads/writes
 - Due to block design
- > Strong data locality (compute runs where data lives)
 - NameNode directs compute to the data
 - Replicas provide multiple places to run task
- > Tight integration with Hadoop ecosystem (YARN, Spark)

When it Doesn't Fit Well

- > When you don't have an Ops team to manage
- > When you need low latency
- > When you have a lot of small files

Where it's Used

HDFS has been replaced by object storage for many workloads

- > Still shows up for jobs running on-prem
- > Useful when you need tight data locality
- > Common when organization runs a Hadoop cluster

Key Takeaways

- > S3 and HDFS were designed for different purposes
- > We saw their tradeoffs around scalability and throughput
- > Increasingly, object stores are used for distributed storage
 - They can be decoupled from compute